Arista Access Node ELEC-ACC-01 Optical Transport Failure SOP

Node: ELEC-ACC-01 | **Vendor:** Arista | **Severity:** Critical | **Alarm Code:** ALARM_1229

Description

The Arista Access Node ELEC-ACC-01 at the Electronic City Hub is equipped with high-speed optical transport interfaces, utilizing Small Form-factor Pluggable (SFP) or Extended Form-factor Pluggable (XFP) transceivers. These transceivers facilitate the transmission and reception of optical signals over single-mode or multimode fibers, respectively. However, the degradation of these transceivers or physical fiber cuts can lead to a critical failure, resulting in a loss of signal integrity and, subsequently, a network disruption. This SOP outlines the procedures to diagnose and resolve the Optical Transport Failure (OTF) on the Arista Access Node ELEC-ACC-01, focusing on the Access role and Arista CLI dialect.

Pre-requisite Checks

- Pre-requisite: Verify that the redundant link is operational and functioning correctly before proceeding with the troubleshooting process.

- Pre-requisite: Ensure that the remote hands are dispatched to the Electronic City Hub to facilitate physical access to the affected node.

- Pre-requisite: Confirm that the Arista Access Node ELEC-ACC-01 is powered on and in a stable state.

Potential Root Causes

Cause	Probability
Degraded SFP/XFP Transceiver	High
Physical Fiber Cut	Medium
Rx/Tx Power Loss	Low

Diagnostic Commands & Expected Outputs

```
Command: show interfaces transceiver, Expected Output: Transceiver Information for Ethernet1/1 SFP-10G-SR, SFP-10G-LR, SFP-100G-CWDM4, etc. Transceiver Type: SFP-10G-SR Transceiver Status: OK Transceiver Temperature: 25.0 C Transceiver Voltage: 3.3 V , Description: This command displays detailed information about the transceiver status, temperature, and voltage, helping to identify any potential issues with the SFP/XFP transceiver.

Command: show interfaces transceiver detail, Expected Output: Transceiver Detail Information for Ethernet1/1 SFP-10G-SR, SFP-10G-LR, SFP-100G-CWDM4, etc. Transceiver Type: SFP-10G-SR Transceiver Status: OK Transceiver Temperature: 25.0 C Transceiver Voltage: 3.3 V Diode Bias Current: 1.5 mA Diode Bias Voltage: 2.5 V , Description: This
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command provides detailed information about the transceiver, including diode bias current and voltage, which can help diagnose issues with the transceiver's optical or electrical components.

Command: show interfaces transceiver stats, Expected Output: Transceiver Statistics for Ethernet1/1 SFP-10G-SR, SFP-10G-LR, SFP-100G-CWDM4, etc. Total Errors: 0 Alignment Errors: 0 FEC Errors: 0 , Description: This command displays statistical information about the transceiver, including total errors, alignment errors, and FEC errors, which can help identify potential issues with the transceiver's optical or electrical components.
```

Resolution Steps

Step 1: Step: Verify the Physical Connection, Command: show interfaces transceiver, Description: Verify the physical connection of the SFP/XFP transceiver to the fiber optic cable and ensure that it is securely connected to the Arista Access Node ELEC-ACC-01. Check for any signs of physical damage or wear on the transceiver or cable.

Step 2: Step: Replace the SFP/XFP Transceiver, Command: no shutdown interface Ethernet1/1, Description: Replace the degraded SFP/XFP transceiver with a new one. Ensure that the new transceiver is compatible with the Arista Access Node ELEC-ACC-01 and the fiber optic cable. Once replaced, apply the 'no shutdown' command to enable the interface.

Step 3: Step: Verify the Fiber Optic Cable, Command: show interfaces transceiver stats, Description: Verify the fiber optic cable for any signs of physical damage or wear. Check the cable's length, connector type, and termination method to ensure that they match the specifications of the Arista Access Node ELEC-ACC-01.

Post-Check Verification

```
Command: show interfaces transceiver, Expected Output: Transceiver Information for Ethernet1/1 SFP-10G-SR, SFP-10G-LR, SFP-100G-CWDM4, etc. Transceiver Type: SFP-10G-SR Transceiver Status: OK Transceiver Temperature: 25.0 C Transceiver Voltage: 3.3 V , Description: Verify that the SFP/XFP transceiver is functioning correctly and that the transceiver status is 'OK'.

Command: show interfaces transceiver stats, Expected Output: Transceiver Statistics for Ethernet1/1 SFP-10G-SR, SFP-10G-LR, SFP-100G-CWDM4, etc. Total Errors: 0 Alignment Errors: 0 FEC Errors: 0 , Description: Verify that the transceiver statistics are accurate and that there are no errors or alignment issues.
```

Roll-back Procedures

- Step: Restore the Original Configuration, Command: copy startup-config running-config, Description: Restore the original configuration of the Arista Access Node ELEC-ACC-01 by copying the startup configuration to the running configuration.

- Step: Revert to the Previous Transceiver, Command: interface Ethernet1/1 no transceiver SFP-10G-SR, Description: Revert to the previous SFP/XFP transceiver by removing the new transceiver and restoring the original one.